Interactions

EES 4760/5760

Agent-Based and Individual-Based Computational Modeling

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Class #14: Tuesday, October 13, 2026



Live web page: https://ees4760.jmgilligan.org/slides/class_14
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Getting Started

- Download files for from the download page for “Interaction Models” on the course web site: ees4760.jmgilligan.org/downloads/14-interaction_class/.

Cooperation and Coordination

Game Theory

- Modern Formal Game Theory originated in the 1940s
  - John von Neumann, *On the Theory of Games of Strategy* (1944)
  - John F. Nash, Jr. “Nash equilibrium”
    - “Equilibrium points in *n*-person games,” Proc. Nat’l. Acad. Sci. **36**, 48 (1950). doi: 10.1073/pnas.36.1.48
    - 1994 Nobel Prize
- Older history (informal)
  - Many centuries of writings on war and gambling
  - Sun Tzu, *The Art of War* (5th century BCE)

> *Knowing the other and knowing oneself: In one hundred battles, no danger.*
> *Not knowing the other and knowing oneself: One victory for one loss.*
> *Not knowing the other and not knowing oneself: In every battle, certain defeat.*

  - Girolamo Cardano, *Liber de Ludo Alea* (*Book on Games of Chance*) (1564)
- Big question:
  - If you are playing a game with another person, what is the best strategy?



John von Neumann (Photo: Los Alamos National Laboratory)



John Nash (Photo: MIT Museum)

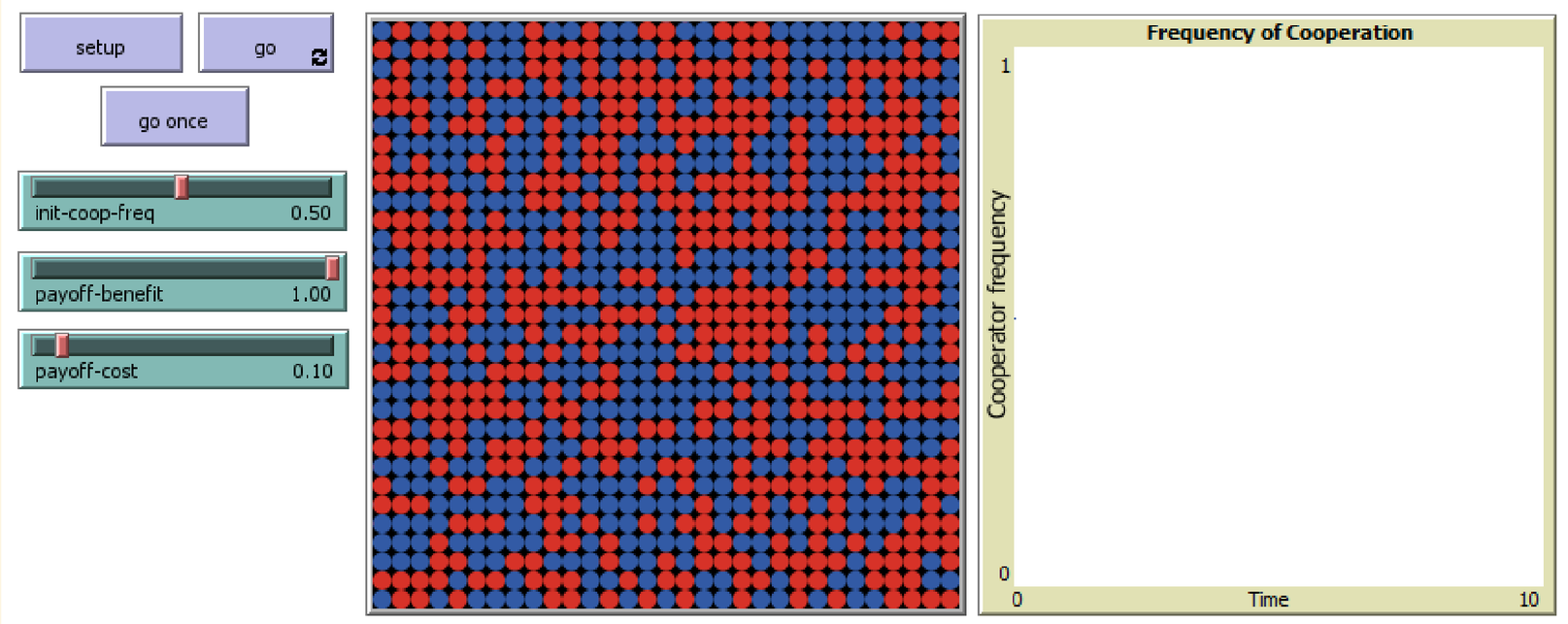
Cooperation vs. Defection (The Prisoner's Dilemma)

- Two people play, and they cannot communicate to discuss their moves.
  - Model:
    - Two people are arrested and accused of a crime
    - If both **cooperate** and remain silent, both are convicted of a minor crime
    - If one **defects** and testifies against the other, they are given a lighter sentence, and the other is convicted of a more serious crime and receives a harsh sentence
    - If **both defect**, they are both convicted of the serious crime, but receive some time off their sentence as a reward for defecting.
  - Mathematical representation:
    - Value for (A, B) of each choice
    - b = benefit of the other person cooperating (lesser crime)
    - c = cost of not defecting (don't get time off the sentence)

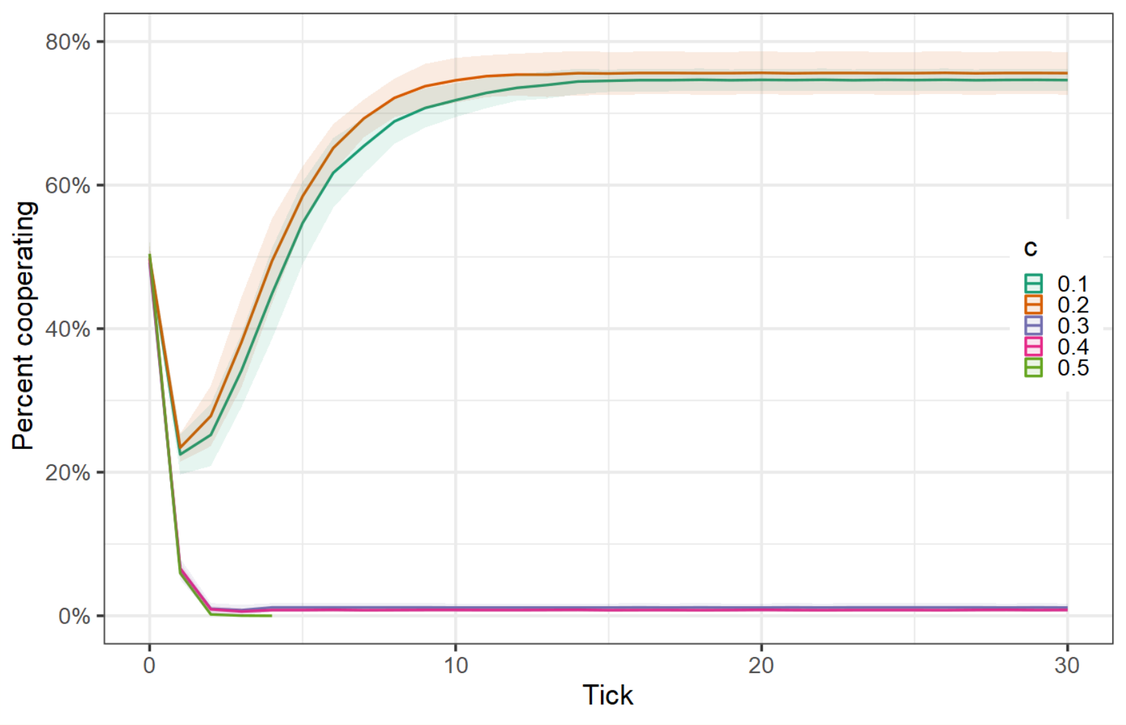
	B cooperates	**B defects**
A cooperates	$b-c, b-c$	$-c, b$
A defects	$b, -c$	0

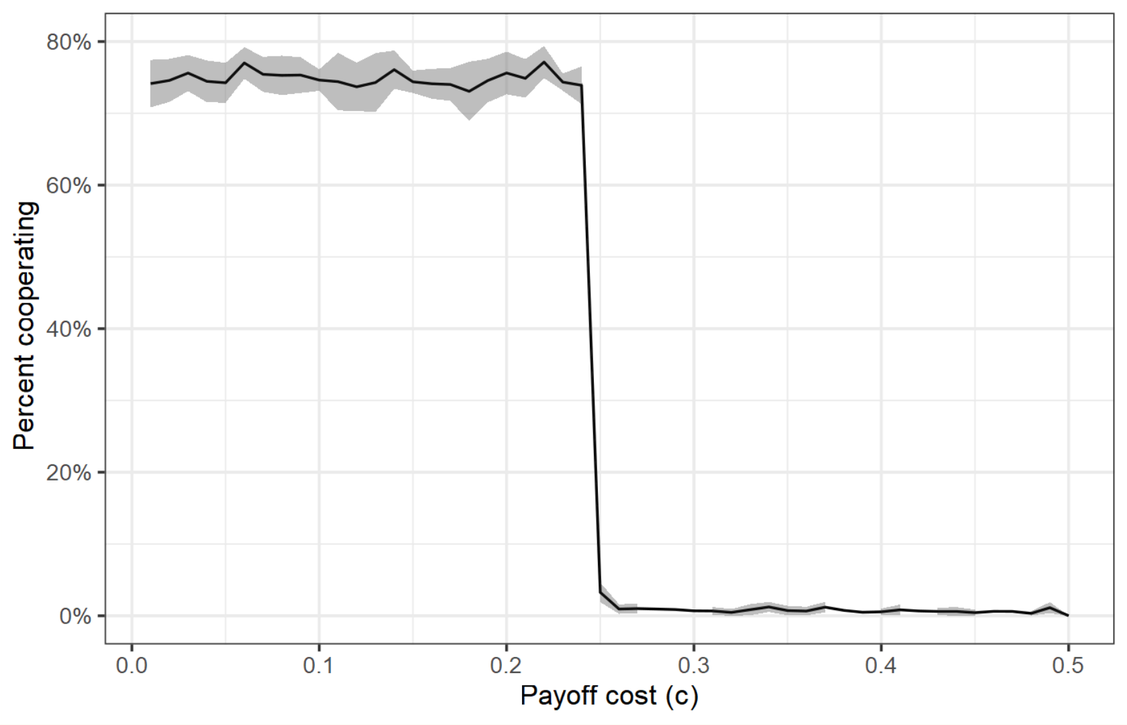
Agent-Based Model: `PD_simple.nlogo`

- Each turtle either always cooperates (blue) or always defects (red)
- Each turtle plays against the four neighbors with which it shares a side
- After each turn the turtle “evolves” by comparing its payoff to its four neighbors and copying the most successful strategy
- What happens when you vary the cost of not defecting (c)?



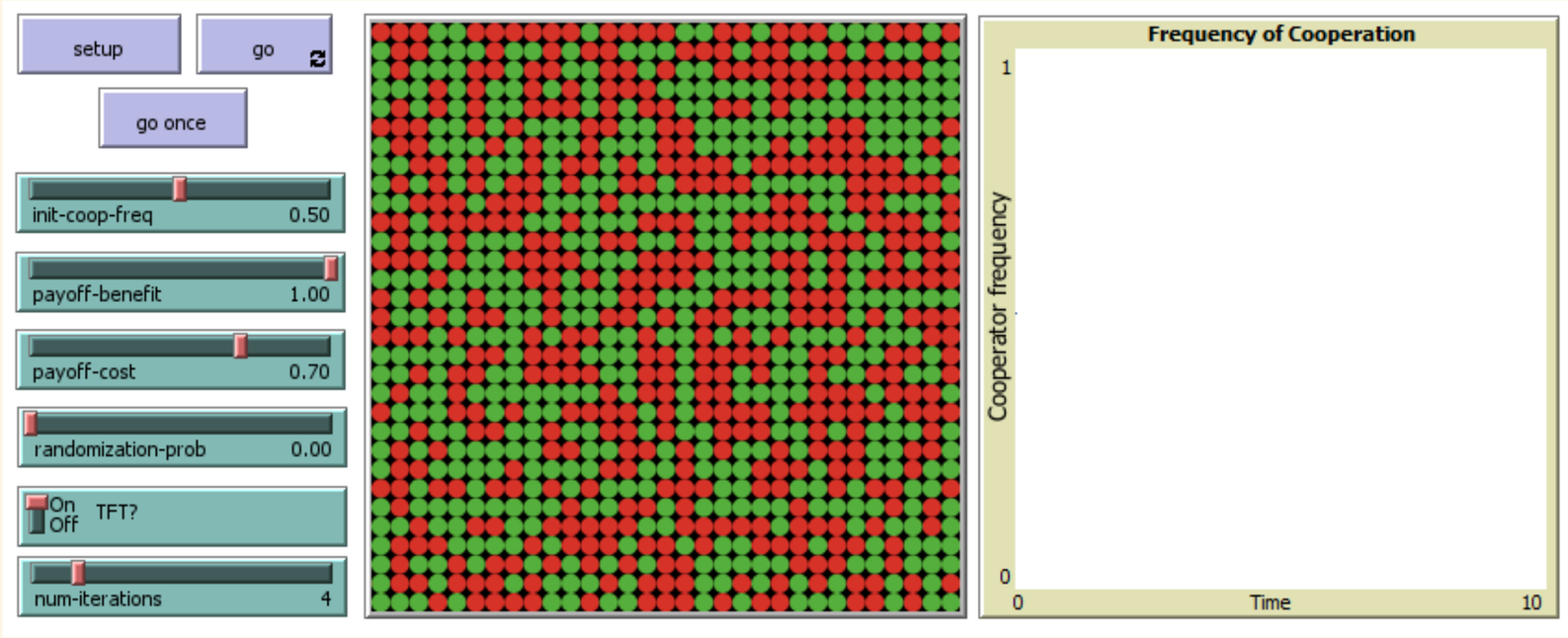
Vary cost of not defecting (*c*)



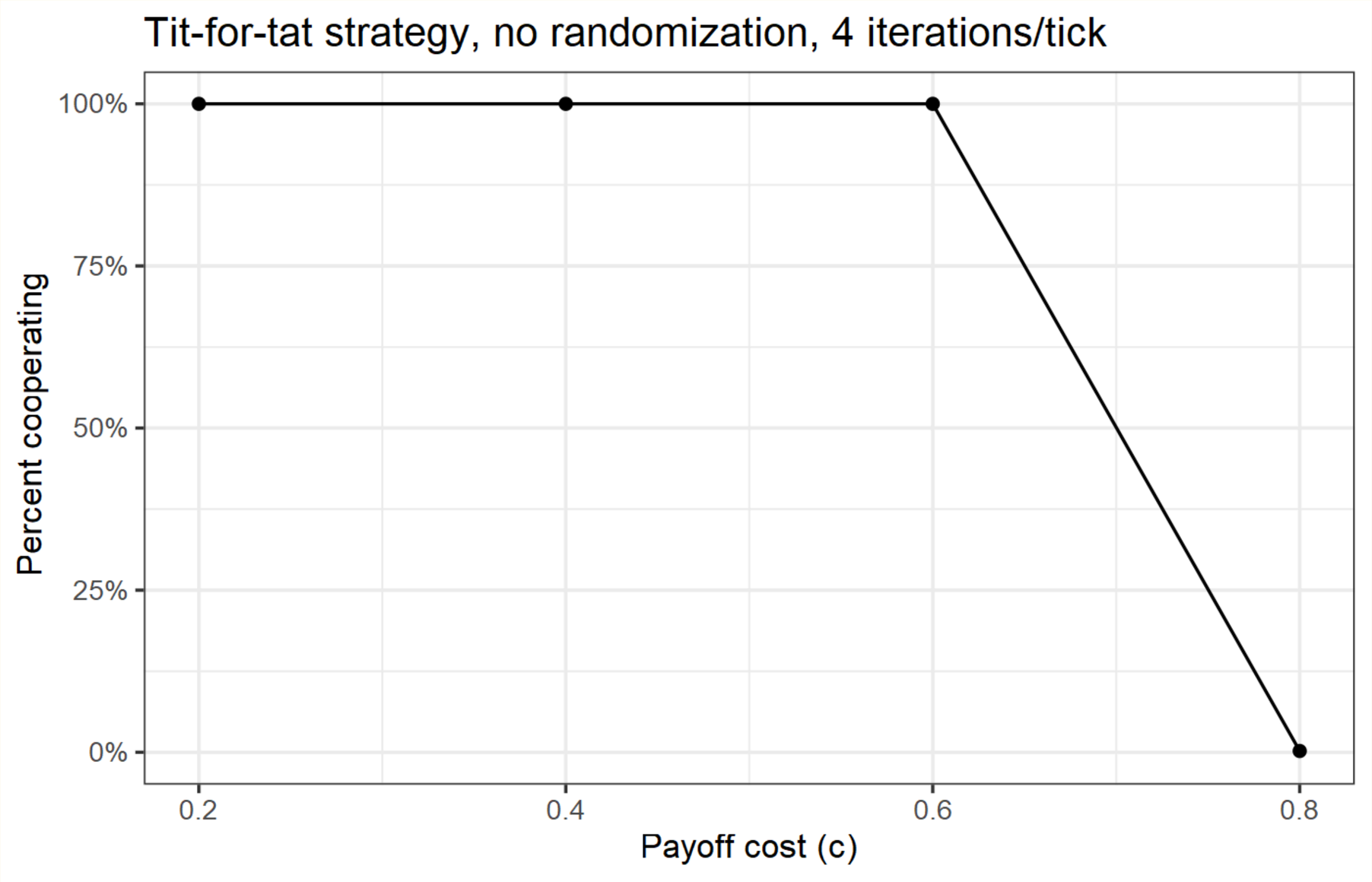


Reciprocity: Tit-for-tat strategy

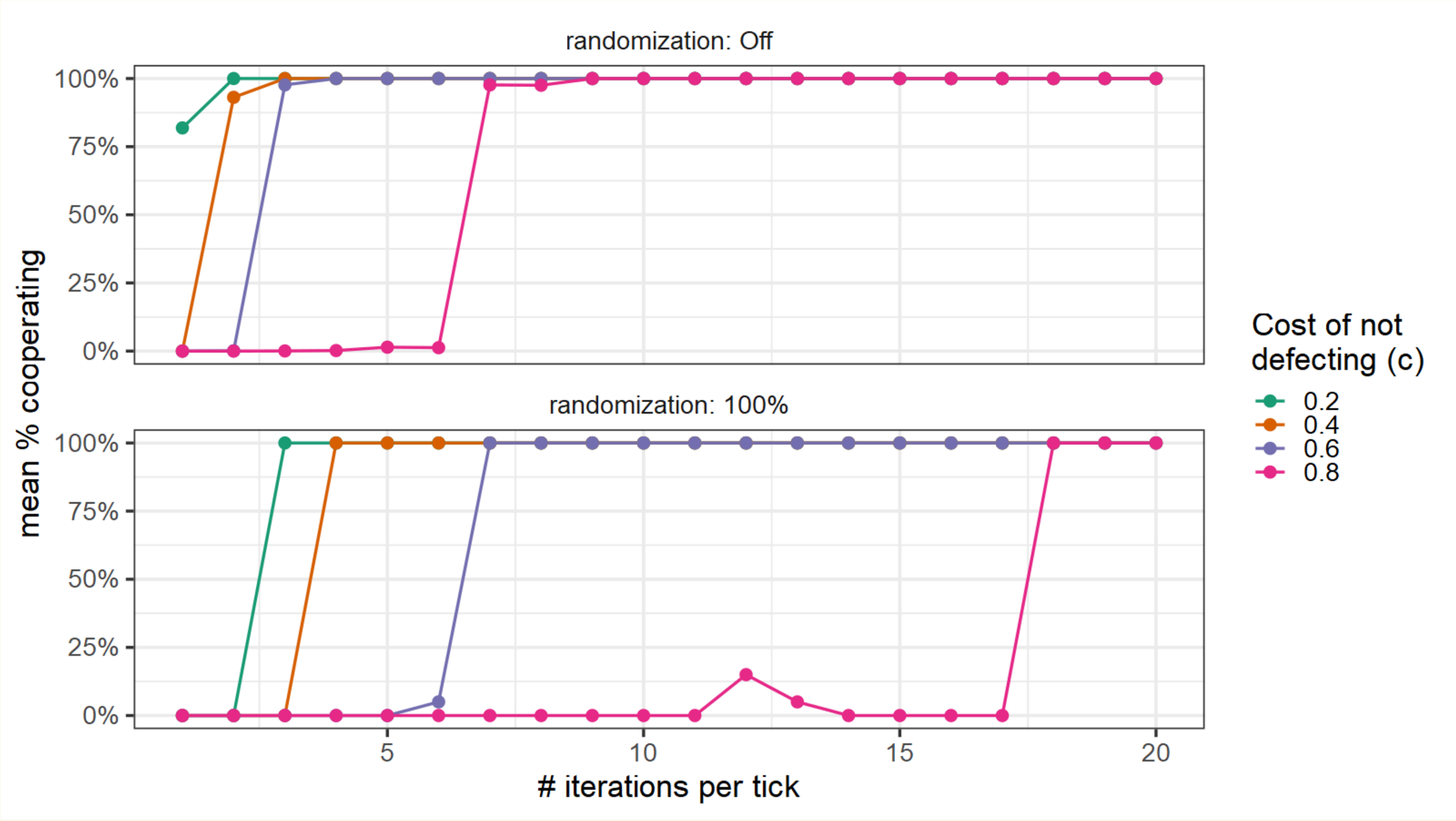
- PD_reciprocity.nlogo
- New strategy: tit-for-tat.
  - Start by cooperating, then copy whatever the opponent did the last time
  - Repeat 4 or more times per tick with each opponent
  - Turtles randomly swap positions each tick, with some probability



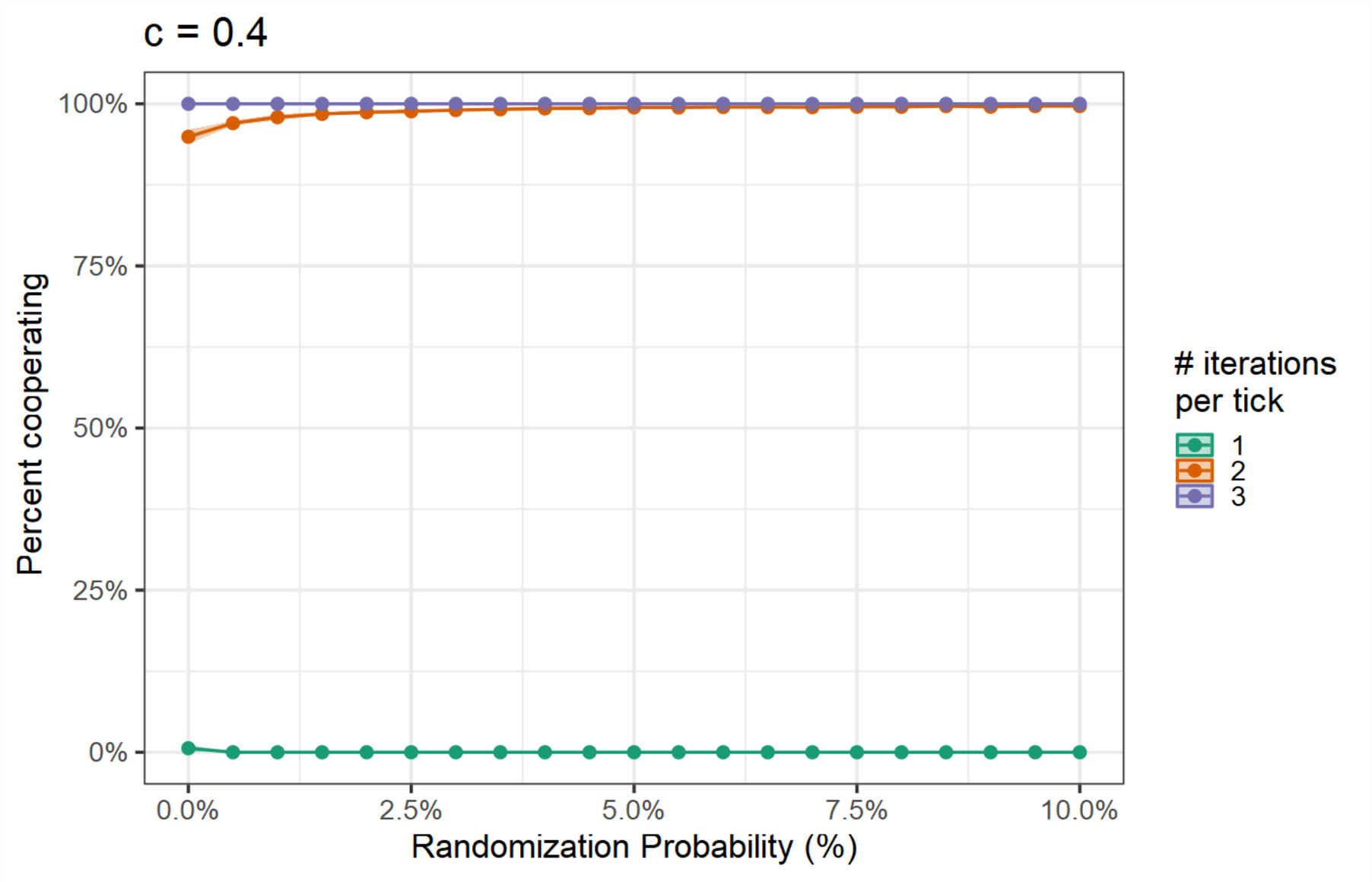
Vary cost of not defecting (*c*)



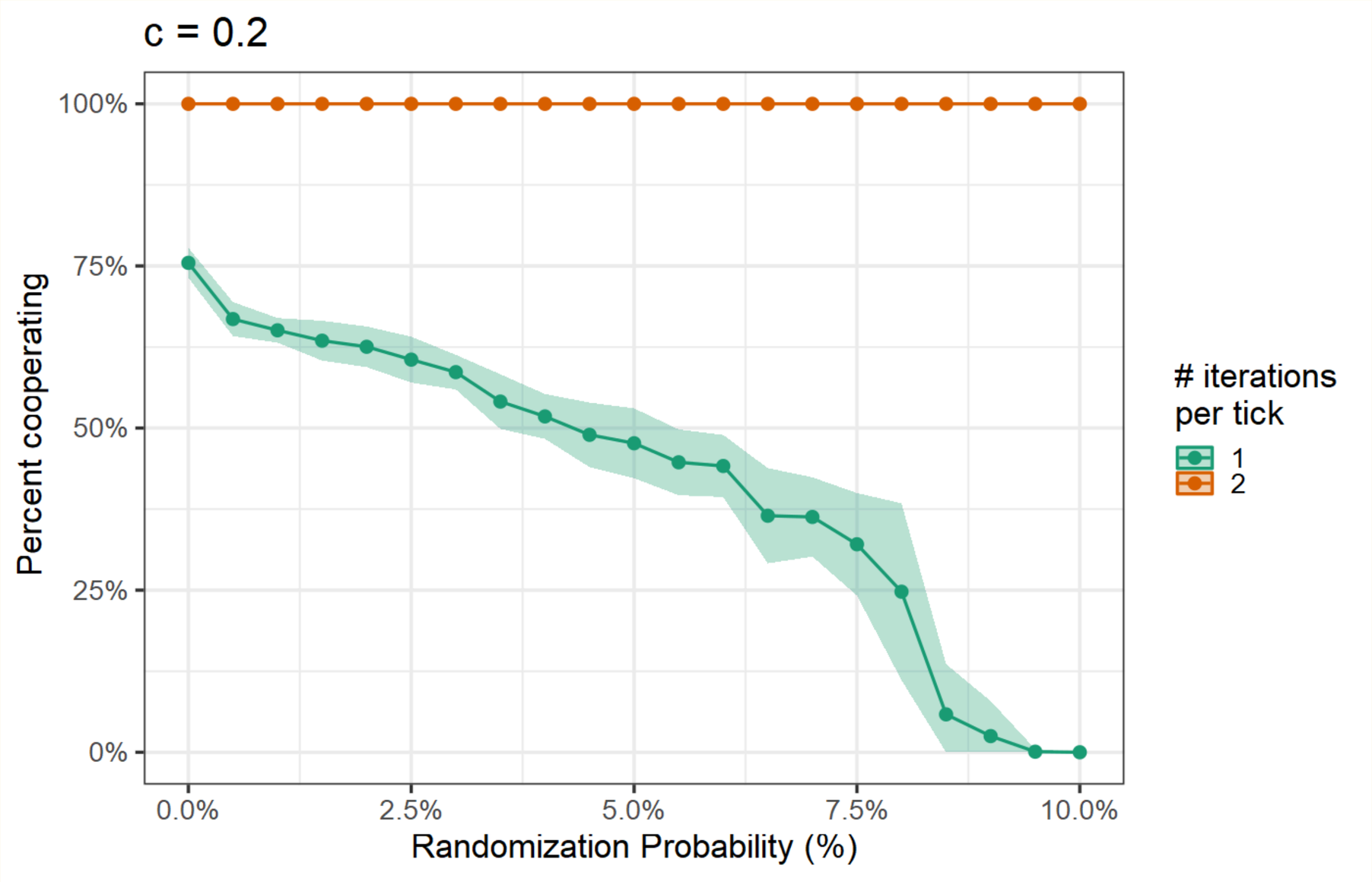
Interaction between randomization and # iterations



Vary randomization



Vary randomization



Work With Partners on Team Projects